

BEYOND REFERENCE MATCHING: SPECIFICATION-BASED EVALUATION OF GENERATED DSP IMPLEMENTATIONS

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ABSTRACT

A reference filter is one valid solution of a design specification, not a definition of the solution set. We evaluate generated DSP implementations by specification-set membership: a candidate is correct if it satisfies a frozen passband, stopband, and stability contract, independently of whether its coefficients match a canonical design. On four FIR/IIR tasks at $f_s = 8$ kHz, nine generated implementations meet the contract while failing coefficient agreement with `firwin` or `butter` (4/4 tasks, 2/3 model families, 6/12 cells). The same contracts accept 12/12 distinct valid controls and reject 12/12 mechanism mutants. First-principles windowed-sinc and frequency-sampling designs occupy the same set without library designers; a tighter mask still admits two distinct same-order occupants. Coefficient and response-to-reference oracles false-reject tone-validated implementations that the specification accepts. Correctness is membership in the feasible set, not agreement with one realization. Our code is available at: <https://github.com/XianghuiMeng-1020/dsp-semantic-correctness>.

Index Terms— Filter design, specification testing, generated implementations, FIR, IIR

1. INTRODUCTION

DSP implementations are increasingly written by program synthesizers and large language models [1, 2]. The default evaluation reflex is to compare the emitted coefficients, or the emitted frequency response, with a trusted reference implementation [3, 4]. That comparison is exact when the specification is a singleton: one correct integer delay, one declared Fourier convention, one alias period. It is not exact for classical filter design.

A magnitude mask with a free transition band defines a *feasible set* of filters, not a unique coefficient vector [5, 6, 7, 8]. Windowed truncation [9, 10], frequency sampling [11], equiripple Parks–McClellan design [12, 13, 14, 15], and IIR Butterworth or elliptic approximations [16, 17, 18, 19, 20] can

all meet the same passband, stopband, and stability bounds while remaining far apart as impulse responses. A canonical library call—Hamming `firwin`, or a Butterworth section—is one occupant of that set.

This paper treats generated DSP code as an evaluation setting for that structure, not as a model leaderboard. The methodological claim is

reference agreement $\neq$ DSP correctness when $|\mathcal{V}_t| > 1$. (1)

We freeze a specification S_t , collect generated and first-principles occupants of $\mathcal{V}_t = \{h : S_t(h) = 1\}$, and compare three oracles: coefficient concordance with one valid reference, constrained-band response similarity to that reference, and specification membership. The last is the correctness object.

2. SPECIFICATION-SET EVALUATION

Let t be a design task and h a realized filter (FIR taps, or an IIR pair (b, a)). A frozen contract returns $S_t(h) \in \{0, 1\}$. The feasible set is

$$\mathcal{V}_t = \{h : S_t(h) = 1\}. \quad (2)$$

Any valid reference h_r satisfies $h_r \in \mathcal{V}_t$ and nothing stronger. Write $\tilde{h}$ for the coefficient vector of h , or the concatenation (b, a) , truncated to the shorter length when orders differ. Coefficient concordance with h_r is

$$R_{t,r}(h) = \mathbf{1} \left[\frac{\|\tilde{h} - \tilde{h}_r\|_2}{\|\tilde{h}_r\|_2} \leq \tau_R \right], \quad (3)$$

with frozen $\tau_R = 0.05$. $R_{t,r}(h) = 1$ asks whether h is the same realization as h_r . $S_t(h) = 1$ asks whether h is an admissible realization of t . These coincide if and only if $\mathcal{V}_t$ is a singleton up to τ_R .

S_t is a magnitude-mask residual on a 4096-point `freqz` grid at $f_s = 8$ kHz. For each constrained band $[f_0, f_1]$ with bounds $[\ell, u]$,

$$\rho = \max_{f \in [f_0, f_1]} \max(\rho_\ell(f), \rho_u(f)), \quad (4)$$

where $\rho_\ell = (\ell - |H|)_+ / (u - \ell)$ and $\rho_u = (|H| - u)_+ / (u - \ell)$ with a 10^{-6} floor on the denominator, and S_t uses the worst ρ over bands. FIR contracts pass at residual $\leq 10^{-6}$; the IIR contract also requires every pole radius < 0.999 and passes at residual $\leq 10^{-3}$. Neither floor was fit to generated residuals. The four frozen masks are

$$\begin{aligned} \text{LP } |H| &\in [0.95, 1.05] \text{ on } [0, 800], \\ |H| &\leq 0.05 \text{ on } [2000, 4000]; \end{aligned} \quad (5)$$

$$\begin{aligned} \text{BP } |H| &\leq 0.06 \text{ on } [0, 500] \cup [3200, 4000], \\ |H| &\in [0.95, 1.05] \text{ on } [1500, 2200]; \end{aligned} \quad (6)$$

$$\begin{aligned} \text{BS } |H| &\in [0.95, 1.05] \text{ on } [0, 600] \cup [3000, 4000], \\ |H| &\leq 0.06 \text{ on } [1400, 2200]; \end{aligned} \quad (7)$$

$$\begin{aligned} \text{IIR } |H| &\in [0.90, 1.10] \text{ on } [0, 600], \\ |H| &\leq 0.10 \text{ on } [2400, 4000], \end{aligned} \quad (8)$$

with frequencies in hertz. Transition bands are unconstrained, as in the classical approximation problem [5, 14]. Linear phase is not required.

Three oracles are scored against a signal-domain battery T : real cosines of length 8192 are passed through `lfilter`, and steady-state Hilbert envelopes are compared with the same mask. Low-pass probes are 50, 400, 800 (pass), 1400 (transition), and 2000, 3000, 3900 Hz (stop); band-pass, band-stop, and IIR probes are the analogous edges and midpoints of (6)–(8). A chirp envelope is not used as a mask residual (valid controls fail that estimator on stopbands).

- **Oracle A** (reference coefficients): $R_{t,r}$.
- **Oracle B** (response similarity): constrained-band $|H|$ RMSE versus h_r , no larger than the maximum same-order control–control RMSE on that task.
- **Oracle C** (specification membership): S_t .

Oracle B is a DSP-native comparison and remains a comparison to *one* element of $\mathcal{V}_t$. Oracle C is membership in $\mathcal{V}_t$.

3. EXPERIMENTAL DESIGN

Generation used Qwen-2.5-72B-Instruct, Llama-3.3-70B-Instruct, and DeepSeek-Chat at temperature 0.7, with four draws per task×model cell. The prompt is the contract statement. No outcome triggered regeneration.

RQ1 — generated implementations. Canonical references are Hamming `firwin` (FIR) and `butter` (IIR) [3]. Before generation, three independently written valid controls per task (`firwin/remez/firls`; `butter/cheby1/ellip`) all satisfy S_t , while every control pair exceeds τ_R . Twelve mechanism mutants (band swap, shifted cutoff, truncated length, unstable pole, Nyquist coded as 1 Hz) all fail S_t . The planned design is $4 \times 3 \times 4 = 48$ generations. Same-order control pairs already realize $|\mathcal{V}_t| > 1$: FIR coefficient ℓ_2

ranges are 0.14–0.27 (LP), 0.26–0.38 (BP), and 0.005–0.051 (BS); IIR (b, a) ℓ_2 ranges are 0.044–0.611. Constrained-band $|H|$ RMSE among those pairs stays below 0.001 (FIR) and 0.019 (IIR); full-grid RMSE is 0.074–0.236 (FIR) because the transition is free. Equal-order FIR controls share group delay.

RQ2 — first-principles occupants. Two textbook FIR constructions are formed with `numpy` only. The Type-I Hamming windowed-sinc low-pass of odd length N and cutoff f_c is

$$h[n] = \frac{w[n]}{\sum_k w[k]} 2f_c \text{sinc}(2f_c(n - M)), \quad (9)$$

$M = (N - 1)/2$, $w[n] = 0.54 - 0.46 \cos(2\pi n/(N - 1))$, f_c in cycles/sample [9, 10]. Band-pass and band-stop designs are complementary differences of (9). Frequency sampling assigns a linear-transition desired magnitude $A[k]$ on the N -point DFT grid, applies the Type-I factor $e^{-j\pi(N-1)k/N}$, and inverts [11]. SciPy design symbols are forbidden in that module. Search is over odd N and cutoff/transition placement until the frozen residual is $\leq 10^{-6}$.

RQ3 — tighter mask. One low-pass analysis mask tightens (5) to $|H| \in [0.99, 1.01]$ on $[0, 800]$ Hz and $|H| \leq 0.01$ on $[1400, 4000]$ Hz (fivefold ripple and stop, twofold narrower free transition). `firwin`, (9), and frequency sampling are compared at a common odd length.

RQ4 — oracle disagreement. On the 14 software-eligible generated implementations, Oracles A–C are scored against T .

4. RESULTS

Counts are occupancy under a frozen protocol, not prevalence.

RQ1. Of 48 requests, 20 executed and 14 passed the type/shape screen. Nine of those 14 satisfy S_t ; all nine fail Oracle A (Table 1). The event occupies 4/4 tasks, 2/3 model families (Llama contributes none), and 6/12 cells. Distinct occupants include a 128-tap `firwin2` low-pass and a 20-tap `remez` low-pass; two 101-tap band-stops with relative $\ell_2 > 0.05$ versus `firwin`; and elliptic (3,3) and Butterworth (4,4) IIR designs, both stable and in-band. Constrained tones through the nine filters are in-mask on 63/63 probes; the twelve valid controls are in-mask on 90/90. On the specification bands, FIR occupants remain close in $|H|$ to the canonical design (RMSE 0.0007–0.0045). Different-order FIR occupants differ in passband group delay by 13.5–30.5 samples. The elliptic pair is not gold-equivalent on the mask: constrained-band $|H|$ RMSE versus Butterworth is 0.065 against a control-pair maximum of 0.019. Two of the nine programs have the same length as `firwin`; one of those two still exceeds τ_R (0.057). A post-hoc swap of h_r among the three frozen controls leaves 7/9 discordant with all three; the two same-length band-stops become concordant with `remez/firls` while remaining discordant with `firwin`.

The swap relocates a program to a different element of $\mathcal{V}_t$, not out of $\mathcal{V}_t$.

RQ2. Both first-principles methods enter $\mathcal{V}_t$ on all three FIR tasks (6/6 shortest designs; Table 2). Locked to the canonical length (81 low-pass, 101 band-pass and band-stop), all six still satisfy S_t ; five of six exceed τ_R versus `firwin`. Hamming windowed-sinc at 101 taps on the band-stop is the same method as `firwin` (coefficient $\ell_2 = 0.0013$, $\Delta\text{GD} = 0$). Frequency sampling at that length remains in $\mathcal{V}_t$ with coefficient ℓ_2 of 0.18–0.39. Every constrained tone is in-mask. Multiplicity is a property of the mask, not of library-API diversity.

RQ3. On the tighter low-pass, the shortest passing lengths are 43 (`firwin` and windowed-sinc; numerically identical) and 57 (frequency sampling). At common length $N = 57$, both distinct methods satisfy the tight contract, pass 7/7 constrained tones, share a 28-sample group delay, and differ by coefficient $\ell_2 = 0.115$ and full-grid $|H|$ RMSE 0.057, with spec-band RMSE 0.0014 (Table 2). A fivefold / twofold tightening of the frozen numbers does not collapse $\mathcal{V}_t$ to a singleton.

RQ4. Table 3 compares oracles with T . Oracle C coincides with T on all 14 eligible implementations (9/9 specification-valid filters are fully in-mask; 5/5 specification failures miss at least one constrained tone). Oracle A agrees on 5/14 and false-rejects every tone-valid implementation ($\text{FRR} = 1$, $\text{FAR} = 0$). Oracle B agrees on 8/14 and still false-rejects 6/9 tone-valid filters ($\text{FRR} = 0.67$), namely the different-order FIR designs and the elliptic pair. The three tone-valid filters that pass Oracle B—two same-order band-stops and the Butterworth (4, 4)—still fail Oracle A. A reference protocol therefore accepts 0 of 48 planned generations and 0/4 tasks; the specification protocol accepts 9 generations and 4/4 tasks. On band-pass and IIR cells, ranking by coefficient distance elevates a specification-failing implementation above a specification-valid one. The band-pass case is a 101-tap `firwin2` with coefficient $\ell_2 = 0.401$ and residual 0.352 (tones 5/9) ranked above a 128-tap specification-valid `firwin2` with $\ell_2 = 1.369$ and residual 0. The IIR case is a specification-failing design with $\ell_2 = 0.499$ versus `butter` ranked above elliptic (3, 3) and Butterworth (4, 4) occupants with residual 0. Kendall τ between coefficient distance and specification residual on the 14 eligible rows is 0.38. Oracle B calibration maxima (control-pair constrained $|H|$ RMSE) are 8.6×10^{-4} (LP), 1.0×10^{-3} (BP), 7.4×10^{-4} (BS), and 1.9×10^{-2} (IIR).

5. CONCLUSION

When a DSP specification admits multiple valid realizations, a reference implementation is one point in $\mathcal{V}_t$. Generated filters, first-principles windowed-sinc and frequency-sampling designs, and a tightened magnitude mask all exhibit the same structure: $S_t = 1$ does not imply coefficient or re-

Table 1. RQ1 occupancy on the frozen specification-set arm. $S_t = 1$ is contract membership; $R_{t,r} = 0$ is coefficient discordance with the canonical reference.

Quantity	Value
Planned / executed / eligible	48 / 20 / 14
$S_t = 1$	9
$S_t = 1$ and $R_{t,r} = 0$	9
Tasks / models / cells	4/4 / 2/3 / 6/12
Pre-generation controls ($S_t = 1$)	12/12
Pre-generation mutants ($S_t = 0$)	12/12
Constrained tones, $S_t = 1$ / controls	63/63 / 90/90

Table 2. RQ2 and RQ3. WS: Hamming windowed-sinc. FS: frequency sampling. Same-order lengths are the frozen `firwin` lengths (RQ2) or $N = 57$ (RQ3).

Setting	In $\mathcal{V}_t$	Same-order	S_t	$\ell_2 > \tau_R$
RQ2 shortest WS/FS (3 FIR tasks)	6/6	—	—	6/6
RQ2 same-order WS/FS	—	6/6	—	5/6
RQ3 tight LP, $N = 57$, WS vs. FS	2/2	2/2	—	1/1 pair

sponse agreement with a single canonical occupant. Coefficient matching and response-to-reference matching remain useful as realization diagnostics. They are not correctness tests. Correctness of a generated DSP implementation is specification-set membership.

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Table 3. RQ4: oracles versus the constrained-tone battery T ($n = 14$ eligible implementations). False rejection is oracle fail and T pass; false acceptance is oracle pass and T fail.

Oracle	Agree with T	FRR	FAR
A (coefficients vs. h_r)	5/14	1.00	0
B (band $ H $ vs. h_r)	8/14	0.67	0
C (specification membership)	14/14	0	0

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